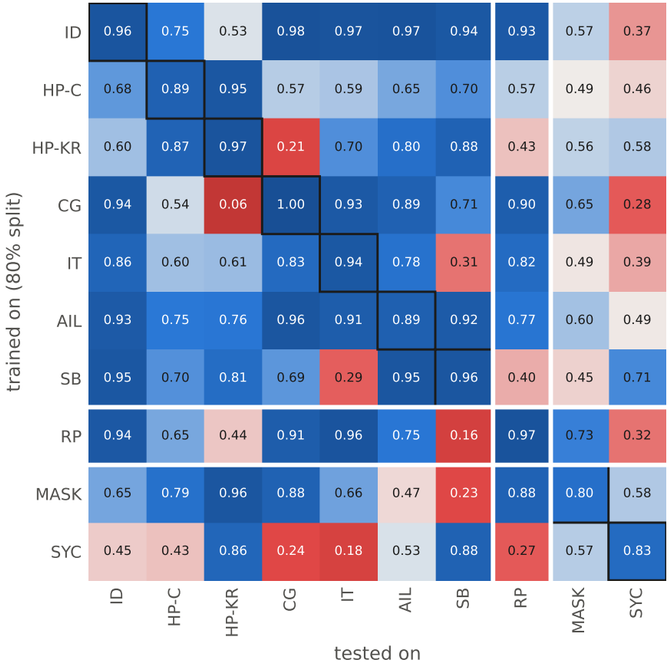
Layer 40 · last pooling · probe: mean-difference (mms) · Llama-3.3-70B-Instruct, 10 datasets · persona bank: paper16 on shared 10+10 questions

Outlined diagonal: tested on the training dataset's own 20% split. Off-diagonal: tested on 100% of the other dataset. White lines separate label constructs (intent | judge | belief).

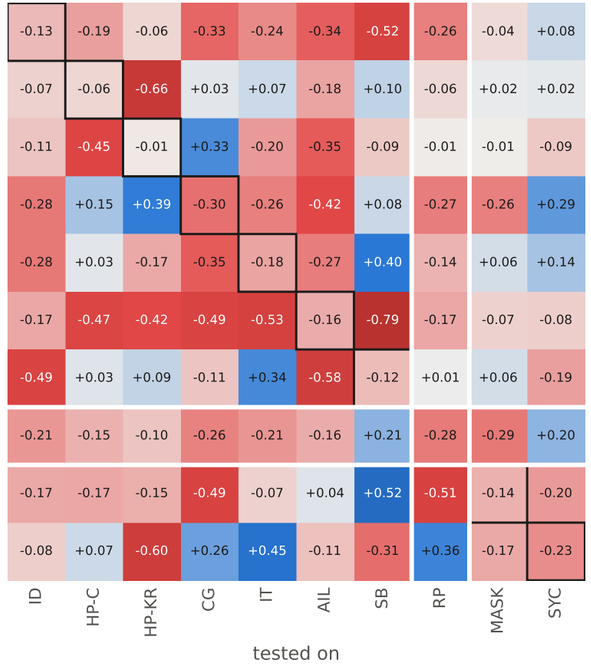
Raw-probe AUROC

diagonal mean 0.920 · off-diagonal mean 0.659



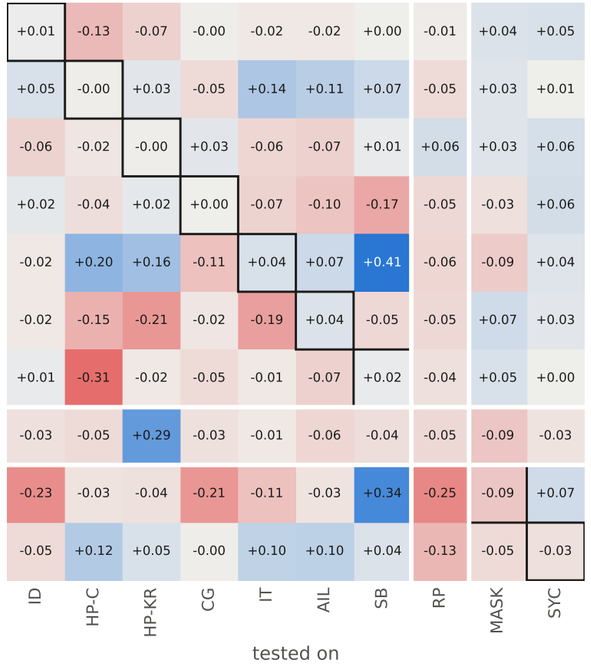
random subspace (k=5) − raw

mean off-diagonal Δ -0.124 · beats raw on 28/90 pairs



dataset PCA (k=5) − raw

mean off-diagonal Δ -0.011 · beats raw on 36/90 pairs



unified-axis PCs (k=5) − raw

mean off-diagonal Δ -0.086 · beats raw on 26/90 pairs

